

Disentangling aleatoric and epistemic risks in bus fleet control: a stable and robust ensemble DRL approach

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Abstract

Bus holding control remains a formidable challenge in urban transit systems, primarily due to the inherent and irreducible stochasticity of traffic conditions and passenger demand, coupled with the difficulty of learning from limited historical data. While Reinforcement Learning (RL) has shown promise, standard actor-critic algorithms frequently suffer from Q -value “poisoning”, where value estimates collapse in highly volatile environments. This instability stems from the entanglement of two distinct uncertainties: aleatoric uncertainty arising from environmental noise, and epistemic uncertainty originating from data insufficiency in the replay buffer.

To circumvent the high sample complexity of direct risk estimation, we draw upon Osgami’s [1] seminal theory on the equivalence between risk-sensitivity and robustness. We propose a robust single-agent RL framework robust-ensemble soft actor-critic (RE-SAC) that reformulates bus fleet control as a dual-risk optimization problem. Our approach explicitly disentangles uncertainties following Clements et al. [2] and Depeweg et al. [3]. To hedge against aleatoric risk, we leverage the connection between robustness and regularization [4, 5], introducing an Integral Probability Metric (IPM)-based weight regularization term [6] that provides a smooth analytical lower bound for the robust Bellman operator. Simultaneously, to handle epistemic risk, we employ a diversified Q -ensemble [7], using predictive variance to penalize over confidence of Q -value estimations in regions with insufficient sample coverage.

From the perspective of robust optimization, our framework offers a novel explanation and solution for the Q -value underestimation phenomenon observed by Ji et al. [8]. We demonstrate that by rigorously managing both data coverage gaps and system stochasticity, RE-SAC prevents the policy from degenerating into the “under-exploitation” trap. Experiments in a realistic bidirectional bus simulation show that our method significantly outperforms multi-agent reinforcement learning (MARL) baselines and vanilla SAC, achieving stable value estimation and superior service regularity across varying noise levels.

1 Introduction

Bus bunching—the phenomenon where multiple buses on the same route cluster together—remains a pervasive challenge in urban transit systems. This instability is driven by a positive feedback loop: a slight delay at a stop increases passenger accumulation, which extends dwell time, further slowing the leading bus while the trailing bus speeds up due to reduced demand [9, 10]. To break this cycle, various holding control strategies have been developed, transitioning from classical schedule-based optimization [11] to sophisticated deep reinforcement learning (DRL) paradigms [12]. However, despite the promising performance of DRL in controlled simulations, these models often exhibit fragile behavior when deployed in realistic, high-variance transit environments.

A critical, yet frequently overlooked bottleneck in applying Actor-Critic methods to transit control is what we term “ Q -value poisoning.” In public transportation, stochasticity is not a transient disturbance but a deeply ingrained feature; fluctuations in passenger arrival rates and road traffic speeds are often irreducible. In such highly volatile environments, standard off-policy algorithms like Soft Actor-Critic (SAC) frequently suffer from pathological value estimation degradation. As

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training progresses, the Q -function may become “poisoned” by environmental noise, leading to severe underestimation of successful past interventions or wild oscillations in unvisited state-action regions. This phenomenon, consistent with the “value collapse” identified by recent studies [8], effectively paralyzes the agent’s policy, causing it to fall into an under-exploitation trap where it fails to maintain regularity despite having encountered optimal trajectories in its experience buffer.

We argue that this instability is deeply rooted in the entanglement of dual uncertainties within the transit system. Following the taxonomy established by Depeweg et al. [3] and Clements et al. [2], we categorize these risks as:

1. **Aleatoric Uncertainty:** The inherent and irreducible randomness of the traffic system, such as stochastic signal timings and bursty passenger surges at stations [13].
2. **Epistemic Uncertainty:** The lack of model knowledge in regions of the state-action space poorly covered by the experience buffer, often exacerbated by the data imbalance inherent in offline or long-tail transit scenarios [7].

In vanilla DRL, these two risks are conflated [14]. An agent might over-penalize a state due to inherent aleatoric noise or show dangerous optimism in epistemic-deficient regions due to function approximation errors [15].

To mitigate Q -value poisoning, a theoretically rigorous approach is to explicitly model the uncertainty distribution using Bayesian methods [16, 3]. However, maintaining full posterior beliefs (e.g., Dirichlet distributions) for every state-action pair is computationally prohibitive in high-dimensional continuous control. To circumvent this, we pivot to the framework of Robust Markov Decision Process (MDP) [17], which seeks a policy optimal against a worst-case uncertainty set. Leveraging the seminal insights of Osogami [1] and Xu et al. [4, 5], we treat this robust objective as mathematically equivalent to regularization in function space. This allows us to efficiently hedge against aleatoric risk by introducing an IPM-based network weight regularization term [6, 18], acting as a structural prior to suppress aleatoric-induced oscillations [19]. Yet, regularization alone is a uniform prior; it cannot detect the “unknown unknowns.” Therefore, for epistemic risk, we complement this with a diversified Q -ensemble [7, 20] to specifically penalize data insufficiency.

Building upon our previous work on single-agent robust control for bus fleets [21], this paper introduces the RE-SAC framework. Our contributions are summarized as follows:

- **Theoretical disentanglement:** We provide a formal motivation for decoupling aleatoric and epistemic risks in transit control, bridging the gap between risk-sensitive RL [1] and bus service reliability.
- **Analytic robustness:** We incorporate an IPM-based weight regularization [6] as a computationally efficient alternative to conservative smoothing [22], preventing Q -value collapse in stochastic environments.
- **Empirical stability:** Through extensive simulations in bidirectional corridors, we demonstrate that our approach prevents the underestimation pitfall [8] and maintains superior service regularity compared to multi-agent baselines.

2 Related work

2.1 Bus fleet control and reinforcement learning

Bus holding control has transitioned from classical theoretical models—such as headway-based exponential models [9] and linear control for schedule reliability [10, 11]—to data-driven optimization strategies. In recent years, DRL has emerged as a powerful tool for dynamic holding decisions [12]. While many studies explore Multi-Agent Reinforcement Learning (MARL) to handle the coupling between vehicles, these systems often struggle with non-stationarity and the high computational cost of coordination [23, 24, 25]. Building upon our previous work, which demonstrated that a single-agent architecture can achieve robust bus fleet control through categorical embeddings [21], this study focuses on the fundamental stability of value estimation in such stochastic environments.

2.2 Uncertainty quantification in RL : From bayesian and risk-sensitivity to robustness

The handling of uncertainty in RL has evolved through two primary lineages: Bayesian distributional modeling and Risk-Sensitive optimization. Bayesian approaches [3, 16] theoretically resolve epistemic uncertainty by maintaining posteriors over system dynamics. Parallel to this, Risk-Sensitive RL explicitly optimizes tail risk measures like Conditional Value at Risk (CVaR) [26, 27], a perspective further advanced by Distributional RL [28, 29]. However, both paradigms face significant hurdles: Bayesian methods struggle with the intractability of posterior updates in high dimensions, while direct risk optimization often incurs prohibitive sampling complexity, as Fei et al. [30] demonstrated that risk-sensitive regret bounds scale significantly worse than standard RL (see Appendix A). To circumvent these computational barriers, the field pivoted towards Robust Markov Decision Processes (RMDPs) [17, 31], where uncertainty is enclosed within a static set. The unification of these views came when Osogami [1] linked robust objectives to risk-sensitive MDPs, bridging worst-case optimization and risk aversion. Further, Geist et al. [32] provided a profound intuition for this connection, showing that regularization in MDPs effectively imposes a *prior distribution* on the policy or value function. Thus, updating with a regularized objective is equivalent to a quasi-Bayesian posterior estimation (see Appendix C), justifying network regularization as an efficient, scalable proxy for handling inherent system stochasticity (Aleatoric Risk).

2.3 Disentangling aleatoric and epistemic risks

Relying solely on regularization acts as a uniform prior, effectively smoothing the policy against aleatoric noise but failing to identify regions of data insufficiency (Epistemic Risk). To address this, we adopt the decomposition framework of Clements et al. [2] and Depeweg et al. [3]. We employ a diversified Q -ensemble [7, 20] to quantify epistemic uncertainty through predictive variance, penalizing over-optimism in sparsely sampled regions. Simultaneously, we use IPM-based weight regularization [6] to enforce local Lipschitz continuity [19], providing a tight lower bound against aleatoric perturbations. This dual approach leverages the efficiency of regularization for aleatoric risk and the sensitivity of ensembles for epistemic risk, preventing the value collapse typically observed in high-variance transit environments [8].

2.4 Stability and the underestimation trap

Function approximation errors are a well-known source of instability in Actor-Critic methods, typically leading to overestimation bias [33, 15]. While SAC utilizes maximum entropy to improve exploration [34], it remains vulnerable to “value collapse” when local batches fail to represent the true high-variance distribution. Recent broad studies suggest that excessive pessimism—often used in offline RL [22, 35]—can lead to an “underestimation trap” in online settings where the agent ignores high-value trajectories [8]. Our work synthesizes these insights to maintain value estimation stability without succumbing to the pathological degradation often seen in vanilla robust implementations [36].

3 Preliminaries

3.1 Maximizing entropy in MDPs

We formulate the bus holding problem as a MDP, characterized by the tuple $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$. In the Maximum Entropy Reinforcement Learning framework [34], the agent’s objective is to jointly maximize the expected cumulative return and the policy’s entropy. This objective is formally expressed as:

$$J(\pi) = \mathbb{E}_{\pi, P} \left[\sum_{t=0}^{\infty} \gamma^t (R(s_t, a_t) + \alpha \mathcal{H}(\pi(\cdot | s_t))) \right], \quad (1)$$

where $\mathcal{H}(\pi(\cdot|s_t))$ denotes the Shannon entropy of the policy, and α is the temperature parameter controlling the exploration-exploitation trade-off. The corresponding soft state value function $V^\pi(s)$ and soft action-value function $Q^\pi(s, a)$ are related by:

$$V^\pi(s) = \mathbb{E}_{a \sim \pi}[Q^\pi(s, a) - \alpha \log \pi(a|s)]. \quad (2)$$

The Soft Bellman Operator $\mathcal{T}^\pi$ acts on Q -functions as $\mathcal{T}^\pi Q(s, a) = R(s, a) + \gamma \mathbb{E}_{s' \sim P}[V^\pi(s')]$, forming a contraction mapping that guarantees convergence to the optimal soft Q -function.

4 Methodology

4.1 Theoretical formulation: Robustness via regularization

To address the inherent stochasticity in transit dynamics, we adopt the RMDP formalism. Instead of assuming a fixed transition kernel P , RMDPs consider an uncertain set of transition probabilities $\mathcal{P}_{s,a}$ for each state-action pair (s, a) [31]. The Robust Bellman Operator $\mathcal{T}^R$ optimizes against the worst-case dynamics within this set:

$$\mathcal{T}^R V(s) = \max_{\pi} \mathbb{E}_{a \sim \pi} \left[R(s, a) + \gamma \inf_{p \in \mathcal{P}_{s,a}} \mathbb{E}_{s' \sim p}[V(s')] \right]. \quad (3)$$

A key challenge in RMDPs is defining a meaningful uncertainty set $\mathcal{P}_{s,a}$. Traditional approaches using KL-divergence balls often yield overly conservative policies. Following recent advances [6], we employ the IPM to define the uncertainty set. The IPM distance between a nominal distribution p° and a perturbed distribution q is defined with respect to a function class $\mathcal{F}$:

$$d_{\mathcal{F}}(p^\circ, q) = \sup_{f \in \mathcal{F}} |\mathbb{E}_{s' \sim p^\circ}[f(s')] - \mathbb{E}_{s' \sim q}[f(s')]|. \quad (4)$$

This formulation allows us to bypass the complexity of finding the worst-case distribution p by directly penalizing the function space complexity, akin to the equivalence between robustness and regularization shown in support vector machines [5] (see Appendix B). When $\mathcal{F}$ is the set of 1-Lipschitz functions, $d_{\mathcal{F}}$ coincides with the 1-Wasserstein distance.

Leveraging the convex duality of IPMs, the inner minimization problem in the Robust Bellman Operator admits a tractable closed-form lower bound. For a value function V_θ parameterized by a neural network, if we constrain the uncertainty set to be a ball $\mathcal{P}_{s,a} = \{q \mid d_{\mathcal{F}}(q, p^\circ) \leq \delta\}$, the robust update becomes:

$$\inf_{q \in \mathcal{P}_{s,a}} \mathbb{E}_q[V_\theta(s')] \approx \mathbb{E}_{p^\circ}[V_\theta(s')] - \delta \cdot \text{Lip}(V_\theta), \quad (5)$$

where $\text{Lip}(V_\theta)$ is the Lipschitz constant of the value network. Since the Lipschitz constant of a deep network is upper-bounded by the product of the spectral norms of its weight matrices regarding each layer l , i.e., $\text{Lip}(V_\theta) \leq \prod_l \|W_l\|_2$, robustness against aleatoric perturbations can be effectively achieved by spectral norm regularization of the critic network weights.

4.2 Disentangling epistemic and aleatoric risks

The fundamental challenge in bus fleet control is that the transition kernel $P(s'|s, a)$ is never truly "nominal." Even with massive historical data, the system faces a dual-layer uncertainty: (i) the inherent traffic stochasticity that exists even in high-data regimes, and (ii) the lack of samples for extreme bunching scenarios or rare traffic incidents. Following the taxonomy of [2, 3], we define the total uncertainty set $\mathcal{P}_{s,a}$ as a composition of Aleatoric and Epistemic components.

In vanilla robust RL, a single uncertainty budget δ is typically used to cover all deviations. However, as observed in our experiments (the "poisoning" of the black line), a monolithic penalty often leads to pathological underestimation [8]. This conflation creates a pathological asymmetry: the agent becomes excessively pessimistic regarding well-explored but stochastic transitions (mistaking

inherent noise for uncertainty) while remaining overly optimistic about unvisited states (due to function approximation bias). High environmental stochasticity exacerbates this misalignment, driving the policy to abandon optimal, high-variance behaviors in favor of effectively random exploration, ultimately triggering value collapse.

To prevent this, we reformulate the Robust Bellman Operator $\mathcal{T}^R$ by decomposing the worst-case expected value into two distinct penalty terms:

$$\mathcal{T}^{REV}(s, a) = R(s, a) + \gamma \left(\mathbb{E}_{s' \sim p^\circ}[V(s')] - \underbrace{\lambda_{epi} \cdot \Gamma_{epi}(s, a)}_{\text{Epistemic Penalty}} - \underbrace{\lambda_{ale} \cdot \Gamma_{ale}(V)}_{\text{Aleatoric Penalty}} \right), \quad (6)$$

where p° is the nominal transition estimated from the experience buffer, and $\lambda_{epi}, \lambda_{ale} > 0$ are coefficients scaling the two types of risk.

1. Epistemic risk (Γ_{epi}): This term accounts for the uncertainty arising from the finite nature of the replay buffer $\mathcal{D}$. In regions where data is sparse (e.g., severe headway deviations), the variance of the value estimate is high. By penalizing this variance, we enforce a "pessimism in the face of the unknown" [7], ensuring the agent remains within the support of the training data.

2. Aleatoric risk (Γ_{ale}): This term accounts for the irreducible noise in the environment (e.g., stochastic passenger arrivals). Based on the IPM duality proven in [6], this risk is captured by the smoothness of the value function V . A non-smooth V with large Lipschitz constants is highly sensitive to aleatoric perturbations. By penalizing the norm of V , we ensure "robustness to the inherent noise."

According to the equivalence theory of Osogami [1] (detailed in Appendix B), this additive decomposition corresponds to an uncertainty set $\mathcal{P}_{s,a}$ that is the intersection of a data-density-based set and a function-class-based IPM set. This disentanglement allows the agent to maintain high Q -values in noisy but well-explored states, thereby avoiding the underestimation trap while maintaining structural robustness.

4.3 Aleatoric risk mitigation via IPM regularization

The aleatoric risk in bus transit arises from the inherent stochasticity of the environment, such as random traffic delays and fluctuating passenger demand. As established in Section 4.1, this risk can be mitigated by bounding the transition uncertainty within an IPM ball, which corresponds to constraining the function class complexity of the value network.

Recall the robust lower bound derived in Eq. (5):

$$\inf_{q \in \mathcal{P}_{s,a}} \mathbb{E}_q[V_\theta(s')] \approx \mathbb{E}_{p^\circ}[V_\theta(s')] - \delta \cdot \text{Lip}(V_\theta). \quad (7)$$

In our deep RL implementation, we approximate the Lipschitz constant $\text{Lip}(V_\theta)$ using the product of spectral norms of the network weights. To make this computationally efficient during training, we simplify the spectral constraint using a Frobenius norm regularization on the weights W of the critic network. Since $\|W\|_2 \leq \|W\|_F$, minimizing the Frobenius norm serves as a computationally efficient upper bound proxy for controlling the Lipschitz constant. The resulting robust update for the Q -function is:

$$\mathcal{T}_{ale}Q(s, a) \approx R(s, a) + \gamma \left(\mathbb{E}_{s' \sim p^\circ}[V(s')] - \delta_{ale} \sum_{l=1}^L \|W_l\|_F^2 \right), \quad (8)$$

where δ_{ale} is the aleatoric uncertainty budget.

Implementation logic in loss function: It is crucial to note the sign of the regularization term. While the robust operator $\mathcal{T}_{ale}$ subtracts the norm term to obtain a pessimistic lower bound, the objective of training the Q -network is to minimize the Mean Squared Bellman Error (MSBE). Therefore, in our loss function $\mathcal{L}(\phi)$, the regularization appears with a positive sign:

$$\mathcal{L}(\phi) = \mathbb{E}_{(s,a) \sim \mathcal{D}} \left[\left(Q_\phi(s, a) - \mathcal{T}_{ale}\hat{Q}(s, a) \right)^2 \right] + \lambda_{ale} \sum_l \|W_l\|_F^2. \quad (9)$$

This positive regularization "resists" the growth of model complexity, forcing the Q -network to remain smooth (low Lipschitz constant), thereby realizing the robust lower bound defined by the IPM uncertainty set. This analytical approach bypasses the need for the expensive inner-loop maximization over state perturbations required by previous methods like RORL [22].

4.4 Epistemic risk mitigation via Q -ensemble

While the IPM-based regularization described in the previous section ensures robustness against inherent environmental noise (aleatoric risk), it is insufficient for handling the distribution shifts inherent in offline-to-online reinforcement learning. In bus transit scenarios, certain states—such as extreme headway deviations or unusual passenger surges—are sparsely represented in the historical data $\mathcal{D}$. In these regions, the value function V_ϕ may produce over-optimistic estimates due to function approximation artifacts rather than true expected rewards. This "lack of knowledge" constitutes the epistemic risk.

To quantify and mitigate this risk, we adopt an ensemble-based pessimism approach [20, 7]. We maintain a set of K independent Q -networks $\{Q_{\phi_k}\}_{k=1}^K$, each initialized randomly. Following the principle that diversified models will agree in high-density data regions but diverge in unvisited regions [7], the variance across the ensemble serves as a proxy for epistemic uncertainty. The epistemic penalty term is defined as:

$$\Gamma_{epi}(s, a) = \text{Var}(\{Q_{\phi_k}(s, a)\}_{k=1}^K) = \frac{1}{K-1} \sum_{k=1}^K (Q_{\phi_k}(s, a) - \bar{Q}(s, a))^2, \quad (10)$$

where $\bar{Q}(s, a)$ is the mean prediction of the ensemble.

In the robust Bellman update, we incorporate this variance to realize "pessimism in the face of the unknown":

$$\mathcal{T}_{epi}Q(s, a) = R(s, a) + \gamma \mathbb{E}_{s' \sim p^o} [\bar{Q}(s', a') - \lambda_{epi} \Gamma_{epi}(s', a')]. \quad (11)$$

By subtracting the variance, the agent is discouraged from selecting actions that lead to states where the ensemble's predictions are inconsistent, effectively constraining the policy to remain within the "trusted" regions of the state space.

Synergy between disentangled uncertainties: The synergy between the Q -ensemble and IPM regularization is the key to preventing Q -value poisoning. As shown in our experiments, using the ensemble alone often fails in high-variance transit environments because it treats aleatoric noise as epistemic gaps, leading to excessive pessimism in well-explored but noisy states. Conversely, using only weight regularization fails to detect OOD regions. By disentangling these two, the ensemble captures the data-density-based risk, while the IPM regularization handles the local-perturbation-based risk. This combined defense ensures that the Q -value remains stable and accurately reflects the worst-case scenario without falling into the underestimation trap identified by [8].

4.5 RE-SAC algorithm implementation

The RE-SAC algorithm integrates the disentangled uncertainty framework into the Soft Actor-Critic (SAC) architecture [34]. To handle the high-dimensional state space and stochastic nature of bus corridors, we maintain an ensemble of K Q -networks $\{Q_{\phi_k}\}_{k=1}^K$ and a policy network π_θ .

4.5.1 Critic objective with disentangled penalties

The core of RE-SAC lies in the robust target calculation. For each transition (s, a, r, s') sampled from the replay buffer $\mathcal{D}$, the robust target y is computed by subtracting the epistemic penalty from the ensemble's mean prediction:

$$y = r + \gamma \left(\mathbb{E}_{a' \sim \pi_\theta} \left[\frac{1}{K} \sum_{k=1}^K \hat{Q}_{\phi_k}(s', a') - \lambda_{epi} \cdot \Gamma_{epi}(s', a') \right] - \alpha \log \pi_\theta(a'|s') \right), \quad (12)$$

where $\hat{Q}_{\phi_k}$ denotes the target Q -networks, and $\Gamma_{epi}(s', a')$ is the epistemic penalty defined as the variance of the ensemble predictions:

$$\Gamma_{epi}(s', a') = \frac{1}{K-1} \sum_{k=1}^K \left(\hat{Q}_{\phi_k}(s', a') - \frac{1}{K} \sum_{j=1}^K \hat{Q}_{\phi_j}(s', a') \right)^2. \quad (13)$$

To mitigate aleatoric risk, we incorporate the IPM-derived weight regularization into the critic loss. The loss function for each Q -network ϕ_k is formulated as:

$$\mathcal{L}_Q(\phi_k) = \mathbb{E}_{(s,a) \sim \mathcal{D}} \left[(Q_{\phi_k}(s, a) - y)^2 \right] + \lambda_{ale} \sum_{l=1}^L \|W_l^{(k)}\|_F^2, \quad (14)$$

where $\|W_l^{(k)}\|_F^2$ is the Frobenius norm of the weights in the l -th layer. It is important to emphasize that the positive sign of the regularization term λ_{ale} ensures that the optimizer penalizes excessive Lipschitz constants, thereby realizing the analytic robust lower bound for aleatoric perturbations. (We provide a guaranteed contraction proof for this operator in Appendix D.5 and a theoretical comparison with SAC’s robustness in Appendix D.)

4.5.2 Actor objective and categorical embeddings

The policy network π_θ is updated to maximize the expected robust value while maintaining sufficient entropy to avoid the underestimation trap identified by Ji et al. [8]:

$$\mathcal{L}_\pi(\theta) = \mathbb{E}_{s \sim \mathcal{D}, a \sim \pi_\theta} \left[\alpha \log \pi_\theta(a|s) - \frac{1}{K} \sum_{k=1}^K Q_{\phi_k}(s, a) \right]. \quad (15)$$

Following our previous work [21], we utilize categorical embeddings to represent discrete transit features. The state s is augmented as:

$$s = [s_{cont} \oplus \text{Emb}(\text{StopID}) \oplus \text{Emb}(\text{BusID})], \quad (16)$$

where $\oplus$ denotes the concatenation of continuous states and d -dimensional embedding vectors. This representation is vital for epistemic uncertainty estimation: since different StopIDs and BusIDs have varying sampling frequencies in $\mathcal{D}$, the ensemble variance Γ_{epi} naturally scales with the data density of specific categorical entities.

4.5.3 Algorithm pseudocode

The complete RE-SAC training procedure is summarized in Algorithm 1.

Algorithm 1 Robust-Ensemble Soft Actor-Critic (RE-SAC)

```
1: Initialize: Ensemble  $Q$ -networks  $\{Q_{\phi_1}, \dots, Q_{\phi_K}\}$ , Actor  $\pi_\theta$ , Replay Buffer  $\mathcal{D}$ .
2: Initialize: Target networks  $\phi'_k \leftarrow \phi_k$ , coefficients  $\lambda_{ale}, \lambda_{epi}$ .
3: for each training episode do
4:   for each environment step  $t$  do
5:     Observe  $s_t$  (including Categorical IDs); Sample  $a_t \sim \pi_\theta(\cdot|s_t)$ .
6:     Execute  $a_t$ , store  $(s_t, a_t, r_t, s_{t+1})$  in  $\mathcal{D}$ .
7:     Sample mini-batch from  $\mathcal{D}$ .
8:     // Critic Update with Disentangled Risks
9:     Compute  $a' \sim \pi_\theta(\cdot|s_{t+1})$ .
10:     $\Gamma_{epi} = \text{Var}(\{Q_{\phi'_k}(s_{t+1}, a')\})$ .
11:     $y = r_t + \gamma \left( \text{Mean}(Q_{\phi'_k}(s_{t+1}, a')) - \lambda_{epi}\Gamma_{epi} - \alpha \log \pi_\theta(a'|s_{t+1}) \right)$ .
12:    for  $k = 1 \dots K$  do
13:      Update  $\phi_k$  by minimizing  $\mathcal{L}_Q(\phi_k)$  with weight  $\text{reg } \lambda_{ale}\|W\|_F^2$ .
14:    end for
15:    // Actor and Temperature Update
16:    Update  $\theta$  by minimizing  $\mathcal{L}_\pi(\theta)$  based on ensemble mean.
17:    Update temperature  $\alpha$  to maintain target entropy  $\mathcal{H}_{target}$ .
18:    // Soft Target Sync
19:     $\phi'_k \leftarrow \tau\phi_k + (1 - \tau)\phi'_k$ .
20:  end for
21: end for
```

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Appendix: Theoretical foundations and derivations

A Sample complexity analysis

We compare the theoretical sample complexity (data efficiency) of Risk-Sensitive RL against our Robust-Regularized approach.

A.1 Risk-Sensitive RL: The exponential barrier

Optimizing risk measures like the Exponential Utility ($\mathbb{E}[\exp(\beta \sum r)]$) faces a fundamental statistical barrier. As established by Fei et al. [30], the regret lower bound for any algorithm learning a risk-sensitive policy is exponential in the risk parameter β and horizon H :

$$\text{Regret}_{RS}(T) \geq \Omega(\exp(|\beta|H) \cdot \sqrt{S^2 AT}). \quad (17)$$

Corollary (Sample complexity): Although Fei et al. prove the regret bound directly, the sample complexity T_ϵ follows as a standard corollary.

Proof Sketch: The average pseudo-regret after T episodes is $\Delta(T) = \text{Regret}(T)/T$. To find an ϵ -optimal policy, we require $\Delta(T) \leq \epsilon$. Substituting the lower bound:

$$\begin{aligned} \frac{\Omega(\exp(|\beta|H)\sqrt{S^2 AT})}{T} &\leq \epsilon \\ \Omega\left(\frac{\exp(|\beta|H)\sqrt{S^2 A}}{\sqrt{T}}\right) &\leq \epsilon \\ \implies \sqrt{T} &\geq \Omega\left(\frac{\exp(|\beta|H)\sqrt{S^2 A}}{\epsilon}\right). \end{aligned} \quad (18)$$

Squaring both sides yields the sample complexity lower bound:

$$T_\epsilon^{RS} \geq \Omega\left(\frac{\exp(2|\beta|H)S^2 A}{\epsilon^2}\right). \quad (19)$$

This confirms that the "Exponential Barrier" is unavoidable for direct risk-sensitive optimization.

A.2 RE-SAC: Retrieving polynomial bounds

Why does RE-SAC achieve polynomial complexity while the equivalent Risk-Sensitive problem is exponential? The key lies in the surrogate objective.

Directly Optimizing $\mathbb{E}[\exp(\beta R)]$ (Risk-Sensitive) requires accurate estimation of the tail distribution, which is statistically expensive (hence $\exp(H)$). However, RE-SAC leverages the specific duality shown by Xu et al. (Appendix B), which maps a specific form of robustness (IPM-based) to Norm Regularization.

$$\text{Robust Objective} \approx \mathbb{E}[R] - \lambda \|W\|_F^2. \quad (20)$$

This transformation is crucial: it converts the "Hard" Risk-Sensitive problem into a "Standard" Risk-Neutral MDP problem where the reward is simply augmented by a deterministic regularizer $\tilde{R}(s, a) = R(s, a) - \lambda \|W\|^2$. Since solving a standard MDP with bounded rewards has polynomial regret $\tilde{O}(\sqrt{H^3 S^2 AT})$, RE-SAC inherits this efficiency. We effectively bypass the need to statistically *estimate* the heavy-tailed distribution (which requires exponential samples) by enforcing a structural constraint (smoothness) that serves as an efficient proxy for risk aversion.

$$\text{Regret}_{RE-SAC}(T) \leq \tilde{O}(\sqrt{H^3 S^2 AT}). \quad (21)$$

Corollary (Sample complexity): Similar to the risk-sensitive case, we derive the sample complexity for RE-SAC. *Proof Sketch:* To ensure the average optimality gap $\Delta(T) \leq \epsilon$:

$$\begin{aligned} \frac{\tilde{O}(\sqrt{H^3 S^2 A T})}{T} &\leq \epsilon \\ \tilde{O}\left(\frac{\sqrt{H^3 S^2 A}}{\sqrt{T}}\right) &\leq \epsilon \\ T &\geq \tilde{O}\left(\frac{H^3 S^2 A}{\epsilon^2}\right). \end{aligned} \quad (22)$$

Remark on lower vs. upper bounds: We compare the *Lower Bound* $\Omega(\cdot)$ of the risk-sensitive problem against the *Upper Bound* $\tilde{O}(\cdot)$ of our approach. For bus control tasks with long horizons ($H \gg 1$) and non-trivial risk aversion ($\beta > 0$), the exponential term dominates the polynomial term:

$$\underbrace{\tilde{O}(\sqrt{H^3})}_{\text{RE-SAC (Poly)}} \ll \underbrace{\Omega(\exp(|\beta|H))}_{\text{Risk-Sensitive (Exp)}}. \quad (23)$$

This confirms Fei et al.'s conclusion [30] that risk sensitivity incurs an unavoidable exponential cost in H , whereas our regularization reduction maintains standard polynomial efficiency.

B The equivalence chain: From risk to regularization

Here we detail the mathematical "Golden Thread" connecting Risk-Sensitivity, Robustness, and Regularization.

B.1 Risk-sensitivity $\iff$ Robustness (Osogami [1])

We provide the rigorous derivation linking Iterated Risk Measures to Robust MDPs. Unlike single-step risk measures, Osogami deals with the dynamic consistency of risk over time.

Definition (Iterated entropic risk): Osogami defines the risk-sensitive value function recursively using the entropic risk measure ρ_β :

$$J_\beta^\pi(s) = r(s, \pi(s)) + \gamma \rho_\beta(J_\beta^\pi(S')), \quad (24)$$

where $\rho_\beta(X) = -\frac{1}{\beta} \log \mathbb{E}[\exp(-\beta X)]$.

Theorem (Bellman equivalence): The Bellman operator for this risk-sensitive objective is $\mathcal{T}_\beta V(s) = \max_a (r + \gamma \rho_\beta(V(s')))$. Using the duality of the log-exponential function (Donsker-Varadhan), we can rewrite the single-step risk as:

$$\rho_\beta(V(s')) = \min_{p \in \Delta(S)} \left(\mathbb{E}_{s' \sim p}[V(s')] + \frac{1}{\beta} \text{KL}(p \| p^\circ(\cdot | s, a)) \right). \quad (25)$$

The term $\frac{1}{\beta} \text{KL}(p \| p^\circ)$ acts as a penalty for deviating from the nominal transition p° . This allows us to rewrite the Risk-Sensitive Bellman operator exactly as a Robust Bellman Operator with a penalty-based uncertainty set:

$$\mathcal{T}_{risk} V(s) \equiv \max_a \min_p \left(\mathbb{E}_p[r + \gamma V(s')] + \frac{\gamma}{\beta} \text{KL}(p \| p^\circ) \right). \quad (26)$$

This proves that optimizing the Iterated Entropic Risk is mathematically isomorphic to solving a Robust MDP with soft KL-constraints.

B.2 Robustness $\iff$ Regularization (Xu et al. [5])

Xu et al. (2009) provide the fundamental link between robust optimization and regularization. They prove this equivalence for Support Vector Machines (Hinge Loss) and Lasso (Regression). We extend their logic to general function approximation.

Theorem (Xu et al.): Consider a supervised learning problem $\min_w \sum_i \ell(y_i, w^T x_i) + \lambda \|w\|$. Xu et al. prove that for specific loss functions, this regularized problem is *equivalent* to a robust formulation $\min_w \max_{\delta \in \mathcal{U}} \sum_i \ell(y_i, w^T (x_i + \delta))$.

- **SVM (Hinge Loss):** Robustness against ℓ_2 -norm perturbation sets $\{\delta : \|\delta\|_2 \leq \epsilon\}$ is equivalent to standard ℓ_2 -regularization (SVM).
- **Lasso (Linear):** Robustness against ℓ_∞ -norm perturbation sets is equivalent to ℓ_1 -regularization (Lasso).

Extension to squared error (Our Case): For the squared error loss $\ell(y, \hat{y}) = (y - \hat{y})^2$ used in Q-learning, adding robustness against ℓ_2 perturbations $\|\delta\|_2 \leq \epsilon$ leads to:

$$\min_w \max_{\|\delta\|_2 \leq \epsilon} (y - w^T(x + \delta))^2 = \min_w (|y - w^T x| + \epsilon \|w\|_2)^2. \quad (27)$$

Expanding this yields:

$$(y - w^T x)^2 + 2\epsilon |y - w^T x| \|w\|_2 + \epsilon^2 \|w\|_2^2. \quad (28)$$

While not identical to simple Ridge Regression $((y - w^T x)^2 + \lambda \|w\|_2^2)$, it strictly enforces a penalty on the weight norm $\|w\|_2$. In RE-SAC, we adopt the standard ℓ_2 regularization $\lambda \|w\|_2^2$ as a computationally efficient surrogate for this robust objective. The core insight remains: imposing limitations on the weight norm (L_2) creates explicit robustness against input (state) perturbations.

B.3 Connection to distributional RL

Algorithms like C51 and QR-DQN explicitly model the return distribution $Z(s, a)$ to manage risk (e.g., optimizing CVaR on Z). While Osogami’s result implies these methods *could* be robust, they require complex distributional projections. Our method uses Xu’s result to bypass distribution modeling entirely, achieving the same robust effect via simple regularization.

C Regularization as a unified semi-bayesian prior

We provide a unified view showing how our Regularized-Robust framework is grounded in Bayesian Uncertainty Quantification.

C.1 The intractability of full bayesian modeling

A fully Bayesian approach to uncertainty would ideally maintain a posterior distribution over the transition dynamics $P(s'|s, a)$. For tabular MDPs, this corresponds to maintaining a Dirichlet Distribution for every state-action pair:

$$P(\cdot|s, a) \sim \text{Dirichlet}(\alpha_1, \alpha_2, \dots, \alpha_S). \quad (29)$$

While theoretically elegant, this "Naive" Bayesian approach faces insurmountable computational hurdles in high-dimensional continuous control (like bus fleets):

- **Explosion of Belief Space:** The state of the system is no longer just the physical state s , but the entire belief state (the parameters α of the Dirichlet). The problem transforms into a POMDP where the belief space dimension is $O(S^2 A)$. For a bus network with $S \approx 10^3$, the parameter space exceeds 10^6 , rendering exact planning intractable.

- **Computational Complexity:** Updating the posterior requires integrating over the simplexes of transition probabilities. As noted by Derman et al., finding the optimal policy in this full Bayesian setting is NP-hard.

This necessitates a tractable approximation that captures the *essence* of Bayesian uncertainty without the computational burden of full belief planning.

C.2 Bayesian robustness via uncertainty decomposition

To make the problem tractable, we adopt a Bayesian Robust approach (Derman et al. [16]), which replaces the complex integration over the posterior with an optimization over "Credible Sets." This first requires creating a tractable posterior proxy, which we achieve via Uncertainty Decomposition (Depeweg et al. [3]).

1. Decomposition (Depeweg et al.): We distinguish between reducible and irreducible uncertainty:

$$\underbrace{\mathcal{H}[p(s'|s, a)]}_{\text{Total Uncertainty}} = \underbrace{\mathbb{E}_{q(\mathbf{w})}[\mathcal{H}(p(s'|s, a, \mathbf{w}))]}_{\text{Aleatoric (Expected Data Noise)}} + \underbrace{I(s'; \mathbf{w})}_{\text{Epistemic (Mutual Information)}}, \quad (30)$$

where $I(s'; \mathbf{w})$ is the Mutual Information between the weights and the prediction.

- **Epistemic Risk:** Corresponds to the spread of the posterior $q(\mathbf{w})$. Instead of full Dirichlet tracking, we approximate this using the Ensemble Variance of Q-functions.
- **Aleatoric Risk:** Corresponds to the inherent noise in the system transition $P(s'|s, a)$. We handle this via Robust Regularization (as proved in Appendix B).

2. Robustness as bayesian proxy (Derman et al.): Derman et al. show that optimizing for the worst-case model within a Bayesian "Credible Set" (Ambiguity Set $\mathcal{U}$) is a lower bound on the true Bayesian optimal value:

$$V_{Bayes}(s) \geq \max_{\pi} \min_{P \in \mathcal{U}_{\alpha}} \mathbb{E}[R]. \quad (31)$$

This justifies our approach: we use the Ensemble to define the Credible Set (where the model might be), and we optimize a lower bound (LCB) to act robustly with respect to this Bayesian uncertainty.

C.3 Regularization as the optimal bayesian update (Geist et al. [32])

Finally, Geist et al. unify these concepts by proving that the Regularized Bellman Operator is not merely a heuristic, but the exact solution to a convex dual problem involving the prior. The regularized Bellman update is defined as:

$$\mathcal{T}_{\Omega} V(s) = \max_{\pi} (\langle Q, \pi \rangle - \Omega(\pi)). \quad (32)$$

Using the Fenchel-Rockafellar duality, the convex conjugate $\Omega^*(Q) = \max_{\pi} (\langle Q, \pi \rangle - \Omega(\pi))$ gives the value function directly. Crucially, if we choose the regularizer $\Omega(\pi)$ to be the KL-divergence from a prior π_0 (i.e., $\Omega(\pi) = \alpha \text{KL}(\pi \| \pi_0)$), the optimal policy update takes the explicit closed form:

$$\pi_{k+1}(a|s) \propto \pi_0(a|s) \exp\left(\frac{Q_k(s, a)}{\alpha}\right). \quad (33)$$

This is exactly a **Bayesian Posterior Update** where:

- π_0 is the **Prior** (historical or base policy).
- $\exp(Q/\alpha)$ is the **Likelihood** of the action being optimal.
- π_{k+1} is the **Posterior** policy.

Geist et al. further prove that this regularized operator $\mathcal{T}_\Omega$ enjoys a strictly tighter contraction factor $\gamma' = \frac{\gamma + \mu\alpha}{1 + \mu\alpha}$ (where μ is the smoothness modulus) compared to the standard operator. **Conclusion:** Our RE-SAC framework is a full-stack Bayesian implementation. The Ensemble captures the Epistemic posterior (Depeweg), the regularized reward handles the Robust Aleatoric risk (Derman/Xu), and the KL-regularized update ensures the policy evolution follows the optimal Bayesian transport (Geist).

D Theoretical comparison: Policy robustness vs. dynamics robustness

We rigorously distinguish the type of robustness provided by standard SAC (Maximum Entropy) versus our RE-SAC (IPM Regularization), demonstrating why the former is insufficient for the dual-uncertainty bus control problem.

D.1 Theoretical fFoundation: The Kantorovich-Rubinstein duality

A core theoretical component of our robust framework is the equivalence between the Wasserstein distance (Optimal Transport) and the IPM over Lipschitz functions. This duality allows us to convert the intractable "Transport Minimal Cost" problem into a tractable "Function Maximization" problem.

Theorem (Kantorovich-Rubinstein Duality [37]): Let P and Q be two probability measures on a metric space (S, d) . The 1-Wasserstein distance is defined as the minimum cost to transport mass from P to Q :

$$W_1(P, Q) = \inf_{\gamma \in \Pi(P, Q)} \mathbb{E}_{(x, y) \sim \gamma} [d(x, y)], \quad (34)$$

where $\Pi(P, Q)$ is the set of all joint distributions (couplings) with marginals P and Q . For the standard cost function $d(x, y) = \|x - y\|$, the Kantorovich-Rubinstein theorem states that this infimum is exactly equal to the supremum over 1-Lipschitz functions:

$$W_1(P, Q) = \sup_{f: \|f\|_L \leq 1} (\mathbb{E}_{x \sim P} [f(x)] - \mathbb{E}_{y \sim Q} [f(y)]). \quad (35)$$

Implication for RE-SAC: In our Robust Bus Control, we define risk as the worst-case transition within a Wasserstein ball of radius ϵ . Using this duality, the robust Bellman regularizer becomes:

$$\max_{\Delta P: W_1(\Delta P, 0) \leq \epsilon} \mathbb{E}_{\Delta P} [V] = \sup_{f: \|f\|_L \leq 1} \epsilon \cdot \|\nabla V\| = \epsilon \cdot \text{Lip}(V). \quad (36)$$

This derivation proves that controlling the Lipschitz constant of the Critic (via Weight Regularization) is mathematically equivalent to minimizing the worst-case parameter shift in the Wasserstein sense.

D.2 SAC: Policy robustness (Eysenbach & Levine, 2021)

Eysenbach and Levine [38] proved that Maximum Entropy RL optimizes a robust objective against adversarial perturbations to the *rewards* or *policy execution*. Specifically, maximizing the entropy-regularized objective:

$$J_{MaxEnt}(\pi) = \mathbb{E}_\pi \left[\sum_t r(s_t, a_t) + \alpha \mathcal{H}(\pi(\cdot | s_t)) \right] \quad (37)$$

is equivalent to maximizing the worst-case reward under an adversarial perturbation $q(a|s)$ that is close to the policy $\pi(a|s)$ in terms of KL divergence:

$$\max_{\pi} \min_{q: D_{KL}(q|\pi) \leq \epsilon} \mathbb{E}_q \left[\sum_t r(s_t, a_t) \right]. \quad (38)$$

Limitation: This form of robustness protects against "Action Noise" (e.g., driver error). However, the derivation relies on Jensen's inequality, providing only a loosely theoretical lower bound. More critically, it assumes fixed system dynamics $P(s'|s, a)$. It does not explicitly penalize sensitivity to state perturbations or model mismatch (Aleatoric/Epistemic risk).

D.3 RE-SAC: Dynamics Robustness (Zhou et al., 2023)

In contrast, our RE-SAC utilizes IPM-based regularization on the Critic, which matches the framework of *Natural Actor-Critic for robust RL* proposed by Zhou et al. [6]. They show that for a Robust MDP with an uncertainty set defined by the IPM (which includes the Wasserstein distance used in transport problems), the robust value function has an exact dual form:

$$V_{Robust}^\pi(s) = \mathcal{T}^\pi V(s) - \delta \|\mathbf{w}\|_p, \quad (39)$$

where $\|\mathbf{w}\|$ is the norm of the critic network weights. Advantage: Unlike the inequality bound in MaxEnt RL, this is an Exact Duality. By penalizing the weight norm, RE-SAC directly restricts the Lipschitz constant of the Q -function with respect to the *state* s :

$$|Q(s, a) - Q(s + \Delta, a)| \leq L_Q \|\Delta\|. \quad (40)$$

This ensures Dynamics Robustness: if the traffic state s jumps unexpectedly (Aleatoric) or drifts to OOD regions (Epistemic), the value estimate Q remains stable.

D.4 Sensitivity analysis conclusion

- **SAC (Policy Robustness):** Smooths the policy $\pi(a|s)$ to handle action execution errors. This defends against driver imperfections but ignores traffic chaos.
- **BEE Operator [8] (Buffer-Based):** Mitigates value collapse by retrieving historical optimal actions from the *external* replay buffer to mechanically "lift" Q -values. While effective, it relies on the coverage of historical success.
- **RE-SAC (Dynamics Robustness):** Stabilizes Q -values *internally* via weight-space regularization. By limiting the sensitivity of Q to state perturbations ($\nabla_s Q$), our method directly addresses the inherent stochasticity of traffic dynamics. This "Dynamics Robustness" is particularly vital for bus fleet control, where the core challenge is the irreducible randomness of the traffic flow ($s \rightarrow s'$), not just the variance in policy execution.

D.5 Theoretical guarantee: Contraction of the RE-SAC operator

A critical question for the proposed RE-SAC Bellman operator is whether it retains the contraction property, which is essential for convergence. The operator $\mathcal{T}^{RE}$ incorporates two penalty terms: the aleatoric regularization (approximating an IPM constraint) and the epistemic ensemble variance (approximating a confidence set constraint). Drawing on the theory of Robust Dynamic Programming by Iyengar [31], we can prove contraction by analyzing the geometry of the uncertainty set.

Iyengar proves that if the uncertainty set $\mathcal{P}_{s,a}$ satisfies the Rectangularity Property (i.e., the uncertainty at each state-action pair is independent), the resulting Robust Bellman Operator is a γ -contraction in the L_∞ norm. Our RE-SAC operator implicitly optimizes over an uncertainty set $\mathcal{P}_{total}$ that is the intersection of two sets:

1. The Aleatoric Set $\mathcal{P}_{ale} = \{p \mid W_1(p, p^\circ) \leq \delta_{ale}\}$, which is (s, a) -rectangular.
2. The Epistemic Set $\mathcal{P}_{epi} = \{p \mid p \in \text{ConfidenceInterval}(D)\}$, which is also constructed to be (s, a) -rectangular.

Since the intersection of two rectangular convex sets is itself a rectangular convex set:

$$\mathcal{P}_{total} = \mathcal{P}_{scale} \cap \mathcal{P}_{epi} \implies \text{Rectangular}. \quad (41)$$

Therefore, $\mathcal{T}^{RE}$ inherits the contraction mapping property from the general Robust MDP framework, guaranteeing a unique fixed point V_{robust}^* .